

Problem 10.8

At time $t = 0$, 1 is deposited into each of Fund X and Fund Y . Fund X accumulates at a force of interest

$$\delta_t = \frac{t^2}{k}.$$

Fund Y accumulates at a nominal rate of discount of 8% per annum convertible semiannually. At time $t = 5$, the accumulated value of Fund X equals the accumulated value of Fund Y . Determine k .

Solution

The accumulated value of Fund X at time 5 is

$$AV_X = \exp \left(\int_0^5 \delta_t dt \right).$$

Substitute $\delta_t = t^2/k$:

$$AV_X = \exp \left(\int_0^5 \frac{t^2}{k} dt \right).$$

Evaluating the integral gives

$$\int_0^5 \frac{t^2}{k} dt = \frac{t^3}{3k} \Big|_0^5 = \frac{125}{3k}.$$

Therefore,

$$AV_X = e^{125/(3k)}.$$

For Fund Y , the nominal discount rate convertible semiannually is

$$d^{(2)} = 0.08.$$

The discount rate per half-year is

$$\frac{0.08}{2} = 0.04.$$

Thus, the accumulation factor per half-year is

$$\frac{1}{1 - 0.04} = \frac{1}{0.96}.$$

There are 10 half-year periods in 5 years, so

$$AV_Y = (0.96)^{-10}.$$

Since the accumulated values are equal,

$$e^{125/(3k)} = (0.96)^{-10}.$$

Taking natural logarithms,

$$\frac{125}{3k} = \ln \left((0.96)^{-10} \right).$$

Solving for k ,

$$k = \frac{125}{3 \ln \left((0.96)^{-10} \right)}.$$

Therefore,

$$\boxed{k = 102.0691594}.$$